

MODULE 5

OSCILLOSCOPES

5.1 Introduction

The Cathode Ray Oscilloscope (CRO) is probably the most versatile tool for the development of electronic circuits and systems. The CRO allows the amplitude of electrical signals, whether they are voltage, current, or power, to be displayed as a function of time. The CRO depends on the movement of an electron beam, which is bombarded (impinged) on a screen coated with a fluorescent material, to produce a visible spot. If the electron beam is deflected on both the conventional axes, i.e. X-axis and Y-axis, a two-dimensional display is produced.

The beam is deflected at a constant rate relative of time along the X-axis and is deflected along the Y-axis in response to an stimulus, such as a voltage. This produces a time-dependent variation of the input voltage. The oscilloscope is basically an electron beam voltmeter. The heart of the oscilloscope is the Cathode Ray Tube (CRT) which makes the applied signal visible by the deflection of a thin beam of electrons. Since the electron has practically no weight, and hence no inertia, therefore the beam of electrons can be moved to follow waveforms varying at a rate of millions of times/second. Thus, the electron beam faithfully follows rapid variations in signal voltage and traces a visible path on the CRT screen. In this way, rapid variations, pulsations or transients are reproduced and the operator can observe the waveform as well as measure amplitude at any instant of time. Since it is completely electronic in nature, the oscilloscope can reproduce HF waves which are too fast for electromechanical devices to follow. Thus, the oscilloscope has simplified many tests and measurements. It can also be used in any field where a parameter can be converted into a proportional voltage for observation, e.g. meteorology, biology, and medicine. The oscilloscope is thus a kind of voltmeter which uses beam instead of a pointer, and kind of recorder which uses an electron beam instead of a pen.

5.2 Basic Principle

5.2.1 Electron Beam

To understand the principle of an oscilloscope, let us consider a torch which is focussed on a piece of cardboard (held perpendicular to the torch). The light beam will make a bright spot where it strikes the cardboard or screen. Hold the torch still, the spot remains still, move the torch, the spot also moves. If the movement is slow, the eye can follow the movement, but if it is too fast for the eye to follow, persistence of vision causes the eye to see the pattern traced by the spot. Hence when we wave the torch from side to side, a horizontal line is traced; we can similarly have a vertical line or a circle. Hence, if the torch is moved in any manner at a very rapid rate, light would be traced, just like drawing or writing. A similar action takes place in the CRT of an oscilloscope. The torch is replaced by an electron gun, the light beam by a narrow electron beam, and the cardboard by the external flat end of a glass tube, which is chemically coated to form a fluorescent screen. Here the electron gun generates the beam which moves down the tube and strikes the screen. The screen glows at the point of collision, producing a bright spot. When the beam is deflected by means of an electric or magnetic field, the spot moves accordingly and traces out a pattern. The electron gun assembly consists of the indirectly heated cathode with its heater, the control grid, and the first and second anodes. The control grid in the CRT is cylindrical, with a small aperture in line with the cathode. The electrons emitted from the cathode emerge from this aperture as a slightly divergent beam. The negative bias voltage applied to the grid, controls the beam current. The intensity (or brightness) of the phosphorescent spot depends on the beam current. Hence this control grid bias knob is called or labelled as intensity. The diverging beam of electrons is converged and focussed on the screen by two accelerating anodes, which form an electronic lens.

Further ahead of the grid cylinder is another narrow cylinder, the first anode. It is kept highly positive with respect to the cathode. The second anode is a wider cylinder following the first. Both the cylinders have narrow apertures in line with the electron beam. The second anode is operated at a still higher positive potential and does most of the acceleration of the beam. The combination of the first anode cylinder and the wider second anode cylinder produces an electric field that focuses the electron beam on the screen, as a lens converges a diverging beam of light. The electronic lens action is controlled by the focus control. If this control is turned to either side of its correct focussing position, the spot on the screen becomes larger and blurred. Bringing it back to its correct position brightens and concentrates the spot. With this proper focus, the small spot can be deflected to produce sharp narrow lines that trace the pattern on the CRT screen. The electron beam may be deflected transversely by means of an electric field (electrostatic deflection) or a magnetic field (electromagnetic deflection). Most oscilloscopes use electrostatic deflection, since it permits high frequency operation and requires negligible power. Electromagnetic deflection is most common in TV picture tubes. Electrons are negatively charged particles; they are attracted by a positive charge or field and repelled by a negative charge. Since the electron beam is a stream of electrons, a positive field will divert it in one direction and a negative field in the opposite direction. To move the beam in this way in the CRT, deflecting plates are mounted inside the tube and suitable deflecting voltages are applied to them. These plates are arranged in two pairs: H_1 and H_2 for deflecting the beam horizontally, and V_1 and V_2 for deflecting it vertically. Leads are taken out for external connections. The beam passes down the tube between the four plates, as shown in Figure 5.1. When the plates are at zero voltage the beam is midway between them and the spot is in the centre of the screen. When H_1 is made positive with respect to the cathode (and all other plates are at zero voltage), it attracts the beam and the spot moves horizontally to the left. When H_2 is made positive, it attracts the beam and the spot moves horizontally to the right. Similarly, when V_1 is made positive, the spot moves vertically upwards and when V_2 is made positive it moves vertically downwards. In each of these deflections, the displacement of the beam, and therefore, the distance travelled by the spot, is proportional to the voltage applied at the plates.

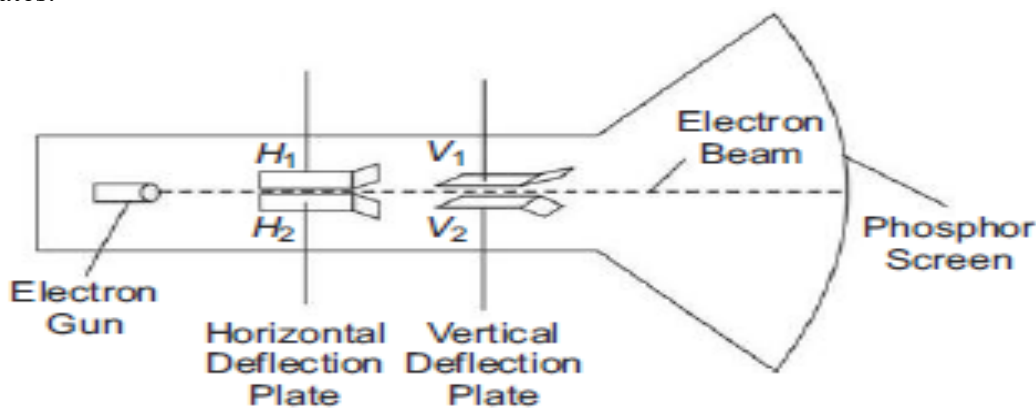


Figure 5.1 Basic Diagram of a CRT

Figure 5.2 shows the various positions of the electron beam for different voltages applied to the two pairs of plates. If a negative voltage is applied to any plate, the beam will be repelled rather than attracted and the deflection will be in the opposite direction. For example, if V_1 is made negative, the beam will be deflected vertically downward. As mentioned before, when a spot moves too rapidly for the eye to follow it traces a line. The same happens when a rapidly pulsating or ac voltage is applied to the deflecting plates, the beam is moved back and forth so rapidly that the spot traces a line. When a positive pulsating voltage is applied to H_1 (or negative pulsating to H_2) the spot traces a horizontal line from the centre to the left.

Similarly, when a positive pulsating voltage is applied to H_2 (or negative to H_1), the spot traces a horizontal line from the centre to the right. Similarly, when the pulsating voltage is applied to V_1 , we get a vertical line from the centre upwards and when applied to V_2 , we get a vertical line from the centre downwards. Now, when an alternating voltage is applied to H_1 or H_2 , the spot moves from the centre to one side, back to the centre and on to the other side, back again and so on, tracing a line that passes through the centre of the screen (because of the attraction and repulsion of the beam by the positive and negative ac half cycles). Hence, a horizontal line is traced when an ac voltage is applied to either horizontal plates. Similarly, a vertical line is traced when an ac voltage is applied to the vertical plates.

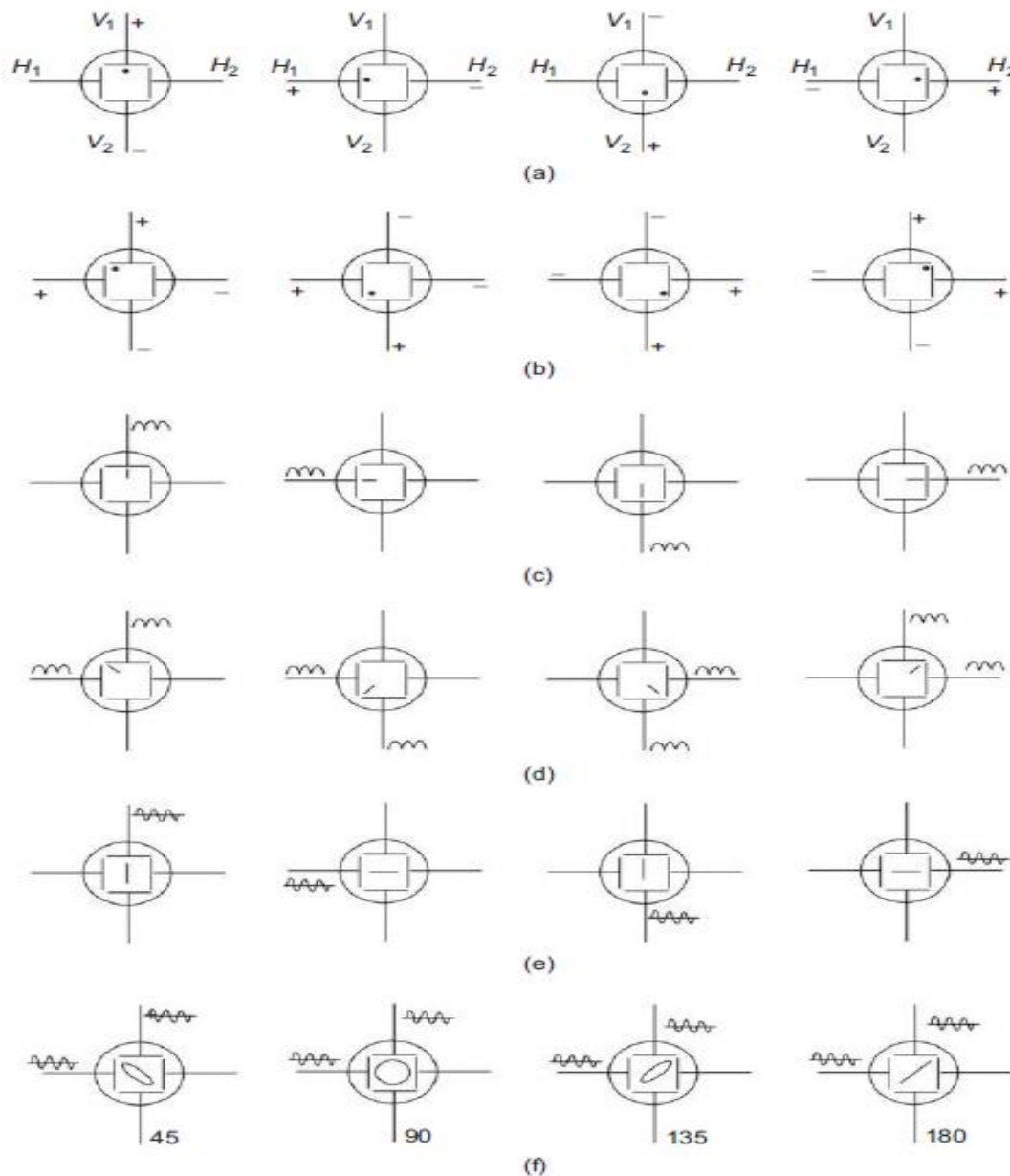


Figure 5.2 (a) Applying dc voltage to vertical and horizontal plates (b) Applying dc voltage to both vertical and horizontal plates (c) Applying pulsating dc to vertical or horizontal plates (d) Applying pulsating dc to both vertical and horizontal (e) Applying a sine wave to vertical or horizontal (f) Applying phase-shifted sine waves to vertical and horizontal plates

Now, let us see what happens to the beam when voltage is applied simultaneously to both vertical and horizontal plates. When voltage is applied to the vertical and horizontal plates simultaneously, the deflection of the beam is proportional to the resultant of the two voltages and the position of the beam is in between the horizontal and vertical axis of the screen. Suppose a steady voltage is applied to one horizontal and one vertical plate. When these two deflection voltages are equal, the position of the spot is 45° . The angle is greater than 45° (spot close to V-axis) when the vertical voltage is greater than the horizontal, and less than 45° (spot close to the H-axis) when the horizontal voltage is greater than the vertical voltage. When the two voltages are reversed in polarity, the deflection is in the opposite direction. If instead of a steady voltage a pulsating positive voltage is applied to the same plates as before, a tilt of 45° is obtained from the horizontal if the two voltages are equal and in phase. The tilt is greater than 45° if the vertical voltage is greater than the horizontal voltage, and less than 45° if the horizontal voltage is greater than the vertical voltage. When a negative voltage is applied to both plates, the trace extends in the opposite direction. When an alternating voltage in phase is applied to the plates, the tilt of the trace is 45° from the horizontal when the two voltages are equal, it traces a straight line at an angle of 45 degrees. Again, the tilt is greater than 45° if the vertical voltage is greater, and less than 45° if the horizontal voltage is greater. The ac trace has equal length from the centre of screen to either tip, when the ac is symmetrical. When it is asymmetrical, the shorter part corresponds to the lower voltage half cycle. A single trace is obtained only when the phase angles are 0° , 180° , or 360° . At other phase angles a double line trace is obtained at equal voltages, the pattern becomes an ellipse with a right tilt for angle between $0-90^\circ$, a circle at 90° and an ellipse with a left tilt between $90-180^\circ$. Again, a left tilt between $180-270^\circ$, a circle at 270° , an ellipse with left tilt between $270-360^\circ$.

5.3 CRT Features

Electrostatic CRTs are available in a number of types and sizes to suit individual requirements. The important features of these tubes are as follows.

1. Size

Size refers to the screen diameter. CRTs for oscilloscopes are available in sizes of 1, 2, 3, 5, and 7 inches. 3 inches is most common for portable instruments. For example, a CRT having a number 5GP1. The first number 5 indicates that it is a 5-inch tube. Both round and rectangular CRTs are found in scopes today. The vertical viewing size is 8 cm and horizontal is 10 cm.

2. Phosphor

The screen is coated with a fluorescent material called phosphor. This material determines the colour and persistence of the trace, both of which are indicated by the phosphor. The trace colours in electrostatic CRTs for oscilloscopes are blue, green and blue green. White is used in TVs, and blue-white, orange, and yellow are used for radar. Persistence is expressed as short, medium and long. This refers to the length of time the trace remains on the screen after the signal has ended. The phosphor of the oscilloscope is designated as follows.

P1 — Green medium

P2 — Blue green medium

P5 — Blue very short

P11 — Blue short

These designations are combined in the tube type number. Hence 5GP1 is a 5-inch tube with a medium persistence green trace. Medium persistence traces are mostly used for general purpose applications. Long persistence traces are used for transients, since they keep the fast transient on the screen for observation after the transient has disappeared. Short persistence is needed for extremely high-speed phenomena, to prevent smearing and interference caused when one image persists and overlaps with the next one. P11 phosphor is considered the best for photographing from the CRT screen.

3. Operating Voltages

The CRT requires a heater voltage of 6.3 volts ac or dc at 600 mA. Several dc voltages are listed below. The voltages vary with the type of tube used.

- (i) Negative grid (control) voltage – 14 V to – 200 V.
- (ii) Positive anode no. 1 (focusing anode) – 100 V to – 1100 V
- (iii) Positive anode no. 2 (accelerating anode) 600 V to 6000 V
- (iv) Positive anode no. 3 (accelerating anode) 200 V to 20000 V in some cases

4. Deflection Voltages

Either ac or dc voltage will deflect the beam. The distance through which the spot moves on the screen is proportional to the dc, or peak ac amplitude. The deflection sensitivity of the tube is usually stated as the dc voltage (or peak ac voltage) required for each cm of deflection of the spot on the screen.

5. Viewing Screen

The viewing screen is the glass face plate, the inside wall of which is coated with phosphor. The viewing screen is a rectangular screen having graticules marked on it. The standard size used nowadays is 8 cm x 10 cm (8 cm on the vertical and 10 cm on horizontal). Each centimeter on the graticule corresponds to one division. The standard phosphor colour used nowadays is blue.

5.4 Block Diagram of Oscilloscope

The major block circuit shown in Figure 5.4, of a general purpose CRO, is as follows:

- 1. CRT
- 2. Vertical amplifier
- 3. Delay line
- 4. Time base
- 5. Horizontal amplifier
- 6. Trigger circuit
- 7. Power supply

The function of the various blocks are as follows.

1. CRT

This is the cathode ray tube which emits electrons that strikes the phosphor screen internally to provide a visual display of signal.

2. Vertical Amplifier

This is a wide band amplifier used to amplify signals in the vertical section.

3. Delay Line

It is used to delay the signal for some time in the vertical sections.

4. Time Base

It is used to generate the sawtooth voltage required to deflect the beam in the horizontal section.

5. Horizontal Amplifier

This is used to amplify the sawtooth voltage before it is applied to horizontal deflection plates.

6. Trigger Circuit

This is used to convert the incoming signal into trigger pulses so that the input signal and the sweep frequency can be synchronised.

7. Power Supply

There are two power supplies, a –ve High Voltage (HV) supply and a +ve Low Voltage (LV) supply. Two voltages are generated in the CRO. The +ve volt supply is from + 300 to 400 V. The –ve high voltage supply is from – 1000 to – 1500 V. This voltage is passed through a bleeder resistor at a few mA. The intermediate voltages are obtained from the bleeder resistor for intensity, focus and positioning controls.

Advantages of using –ve HV Supply:

- (i) The accelerating anodes and the deflection plates are close to ground potential. The ground potential protects the operator from HV shocks when making connections to the plates.
- (ii) The deflection voltages are measured wrt ground, therefore HV blocking or coupling capacitors are not needed, but low voltage rating capacitors can be used for connecting the HV supply to the vertical and horizontal amplifiers.
- (iii) Less insulation is needed between positioning controls and chassis.

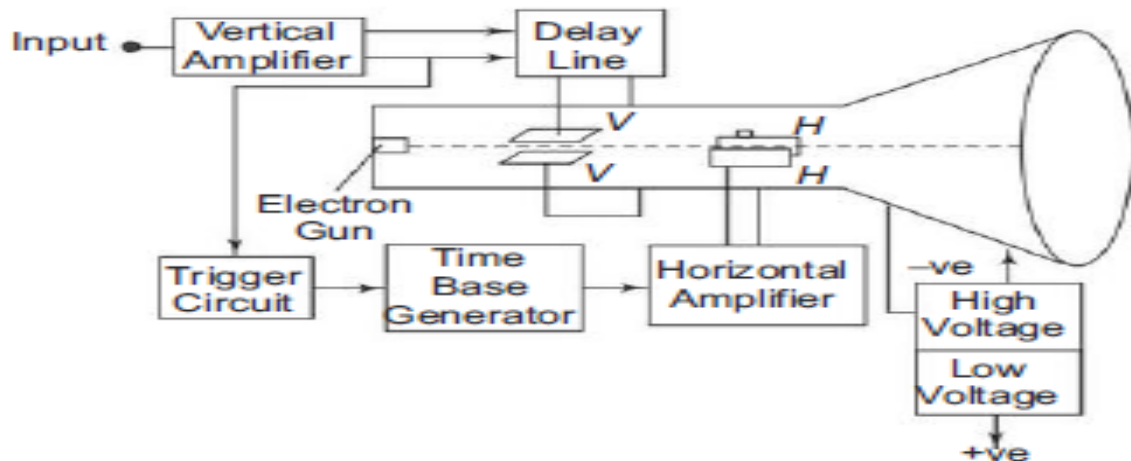


Figure 5.4 Basic CRO block diagram

5.5 Dual Beam CRO

Figure 5.5 illustrates a block diagram of a Dual Beam CRO. The dual trace oscilloscope has one cathode ray gun, and an electronic switch which switches two signals to a single vertical amplifier. The dual beam CRO uses two completely separate electron beams, two sets of VDPs and a single set of HDPs. Only one beam can be synchronised at one time, since the sweep is the same for both signals, i.e. a common time base is used for both beams. Therefore, the signals must have the same frequency or must be related harmonically, in order to obtain both beams locked on the CRT screen, e.g. the input signal of an amplifier can be used as signal A and its output signal as signal B.

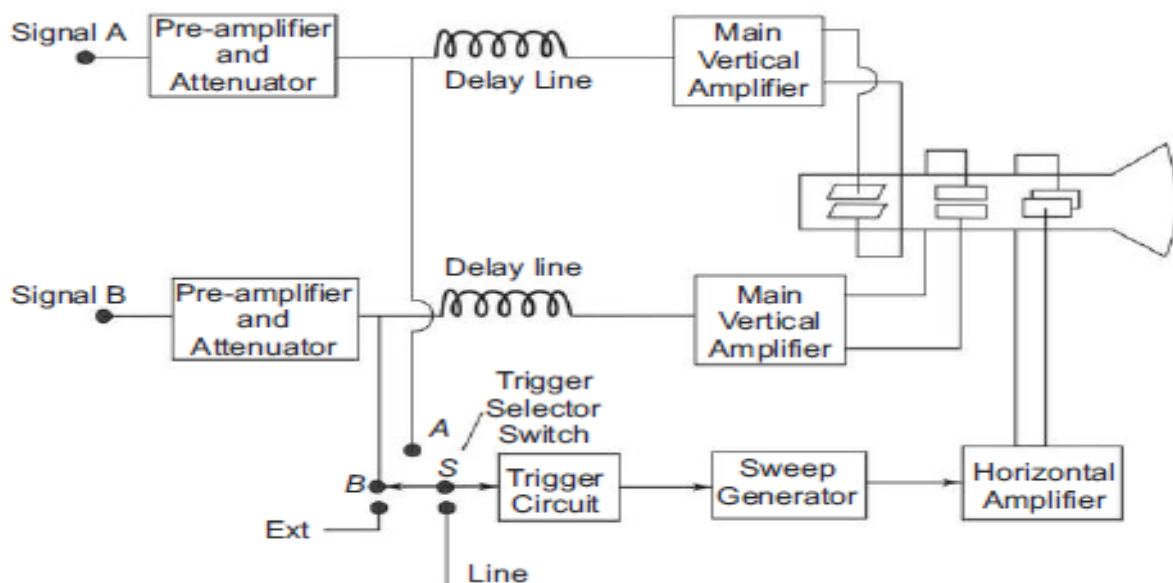


Figure 5.5 Dual beam CRO

5.6 Dual Trace Oscilloscope

Figure 5.6 (a) shows a block diagram of a dual trace oscilloscope. This CRO has a single electron gun whose electron beam is split into two by an electronic switch. There is one control for focus and another for intensity. Two signals are displayed simultaneously. The signals pass through identical vertical channels or vertical amplifiers. Each channel has its own calibrated input attenuator and positioning control, so that the amplitude of each signal can be independently adjusted.

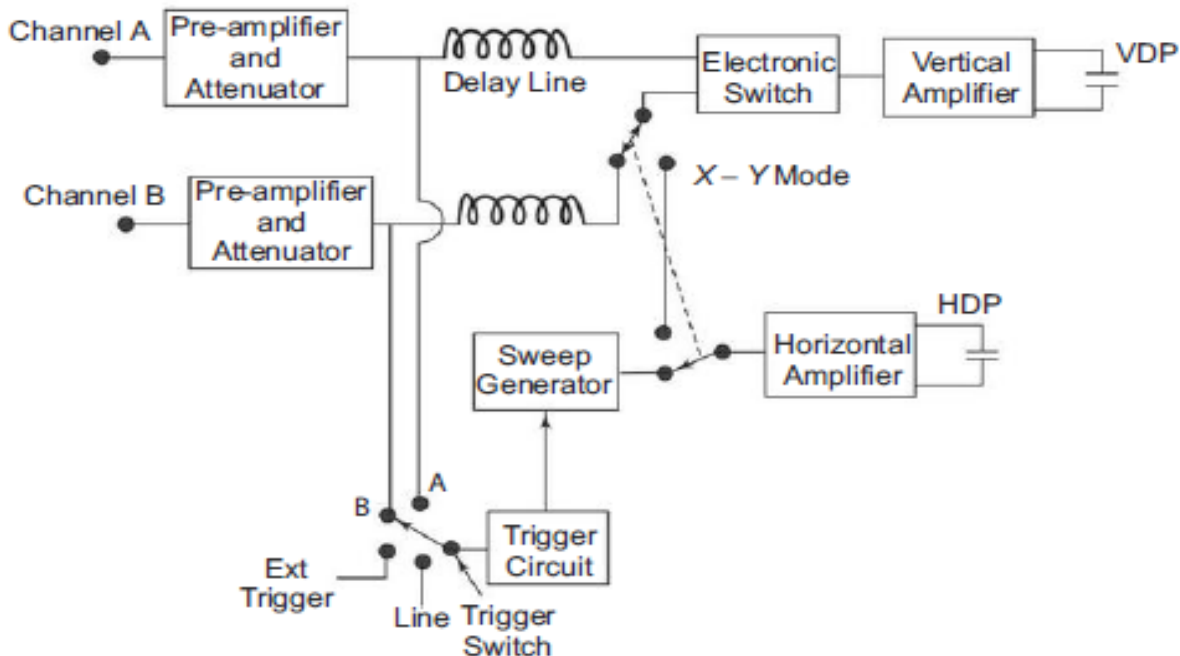


Figure 5.6 (a) Dual trace oscilloscope

A mode control switch enables the electronic switch to operate in two modes i.e. Alternate and Chop mode. When the switch is in ALTERNATE position, the electronic switch feeds each signal alternately to the vertical amplifier. The electronic switch alternately connects the main vertical amplifier to channels A and B and adds a different dc component to each signal; this dc component directs the beam alternately to the upper or lower half of the screen. The switching takes place at the start of each new sweep of the sweep generator. The switching rate of the electronic switch is synchronised to the sweep rate, so that the CRT spot traces the channel A signal on one sweep and the channel B signal on the succeeding sweep (Figure 5.6 (b)).

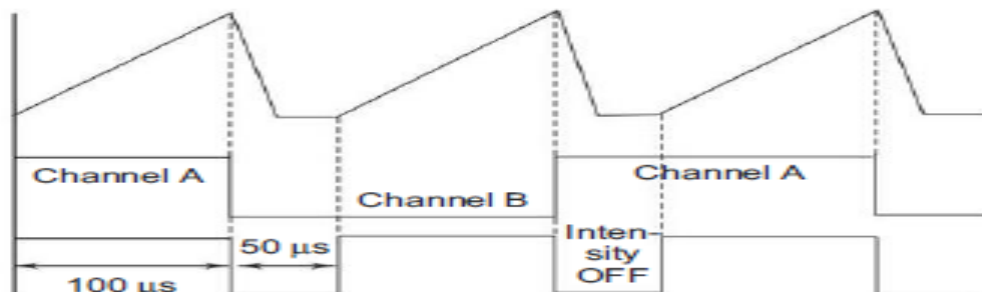


Figure 5.6 (b) Time relation of a dual-channel vertical amplifier in alternate mode

The sweep trigger signal is available from channels A or B and the trigger pick-off takes place before the electronic switch. This arrangement maintains the correct phase relationship between signals A and B.

When the switch is in the CHOP mode position, the electronic switch is free running at the rate of 100–500 kHz, entirely independent of the frequency of the sweep generator. The switch successively connects small segments of A and B waveforms to the main vertical amplifier at a relatively fast chopping rate of 500 kHz e.g. 1ms segments of each waveform are fed to the CRT display (Figure 5.6 (c)).

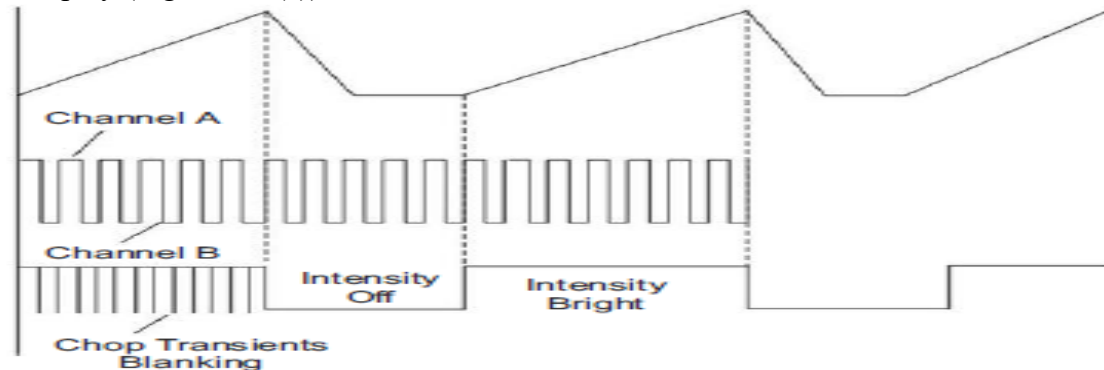


Figure 5.6 (c) Time relation of a dual-channel vertical amplifier in chop mode

If the chopping rate is slow, the continuity of the display is lost and it is better to use the alternate mode of operation. In the added mode of operation a single image can be displayed by the addition of signal from channels A and B, i.e. $(A + B)$, etc. In the X – Y mode of operation, the sweep generator is disconnected and channel B is connected to the horizontal amplifier. Since both pre-amplifiers are identical and have the same delay time, accurate X – Y measurements can be made.

5.7 Storage Oscilloscope (For VLF Signal)

Storage targets can be distinguished from standard phosphor targets by their ability to retain a waveform pattern for a long time, independent of phosphor persistence. Two storage techniques are used in oscilloscope CRTs, mesh storage and phosphor storage. A mesh-storage CRT uses a dielectric material deposited on a storage mesh as the storage target. This mesh is placed between the deflection plates and the standard phosphor target in the CRT. The writing beam, which is the focussed electron beam of the standard CRT, charges the dielectric material positively where hit. The storage target is then bombarded with low velocity electrons from a flood gun and the positively charged areas of the storage target allow these electrons to pass through to the standard phosphor target and thereby reproduce the stored image on the screen. Thus, the mesh storage has both a storage target and a phosphor display target. The phosphor storage CRT uses a thin layer of phosphor to serve both as the storage and the display element. Mesh Storage It is used to display Very Low Frequencies (VLF) signals and finds many applications in mechanical and biomedical fields. The conventional scope has a display with a phosphor persistence ranging from a few microseconds to a few seconds. The persistence can be increased to a few hours from a few seconds.

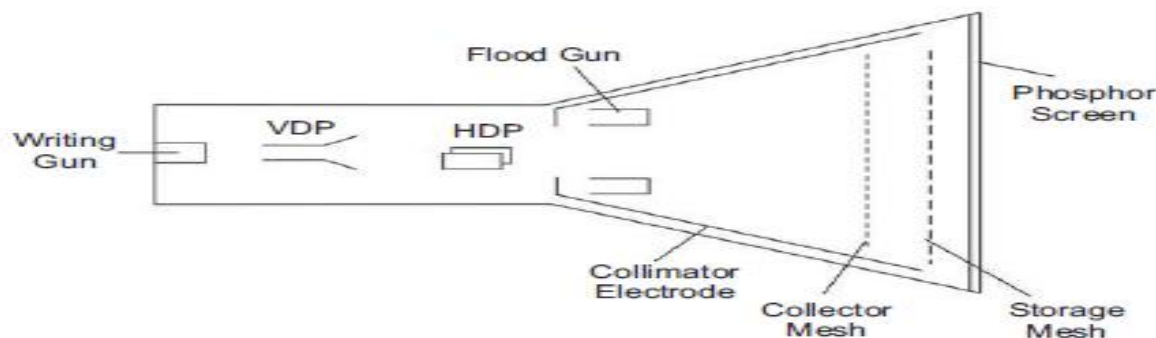


Figure 5.7 (a) Basic elements of storage mesh CRT

A mesh storage CRT, shown in Figure 5.7 (a), contains a dielectric material deposited on a storage mesh, a collector mesh, flood guns and a collimator, in addition to all the elements of a standard CRT. The storage target, a thin deposition of dielectric material such as Magnesium Fluoride on the storage mesh, makes use of a property known as secondary emission. The writing gun etches a positively charged pattern on the storage mesh or target by knocking off secondary emission electrons. Because of the excellent insulating property of the Magnesium Fluoride coating, this positively charged pattern remains exactly in the position where it is deposited. In order to make a pattern visible, a special electron gun, called the flood gun, is switched on (even after many hours). The electron paths are adjusted by the collimator electrode, which constitutes a low voltage electrostatic lens system (to focus the electron beam), as shown in Figure 5.7 (b).

Most of the electrons are stopped and collected by the collector mesh. Only electrons near the stored positive charge are pulled to the storage target with sufficient force to hit the phosphor screen. The CRT will now display the signal and it will remain visible as long as the flood guns operate. To erase the pattern on the storage mesh, a negative voltage is applied to neutralise the stored positive charge. Since the storage mesh makes use of secondary emission, between the first and second crossover more electrons are emitted than are absorbed by the material, and hence a net positive charge results.

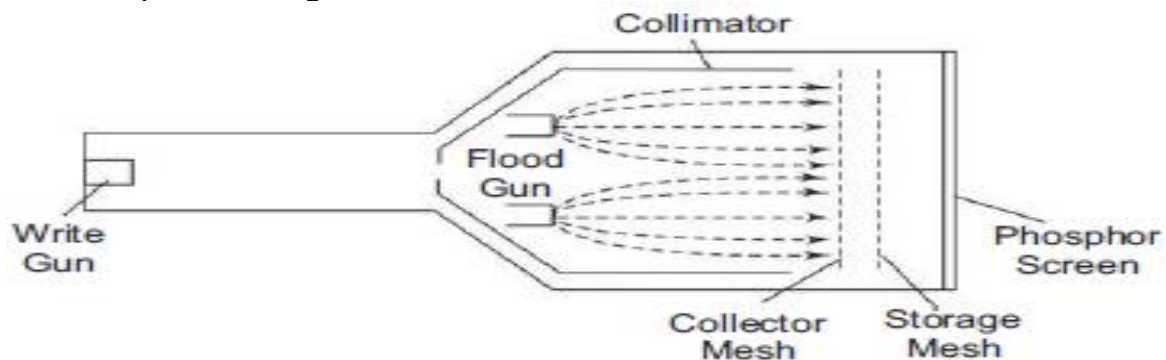


Figure 5.7 (b) Storage mesh CRT

Below the first crossover a net negative charge results, since the impinging electrons do not have sufficient energy to force an equal number to be emitted. In order to store a trace, assume that the storage surface is uniformly charged and write gun (beam emission gun) will hit the storage target. Those areas of the storage surface hit by the deflecting beam lose electrons, which are collected by the collector mesh. Hence, the write beam deflection pattern is traced on the storage surface as a positive charge pattern. Since the insulation of the dielectric material is high enough to prevent any loss of charge for a considerable length of time, the pattern is stored. To view, the stored trace, a flood gun is used when the write gun is turned off. The flood gun, biased very near the storage mesh potential, emits a flood of electrons which move towards the collector mesh, since it is biased slightly more positive than the deflection region. The collimator, a conductive coating on the CRT envelope with an applied potential, helps to align the flood electrons so that they approach the storage target perpendicularly. When the electrons penetrate beyond the collector mesh, they encounter either a positively charged region on the storage surface or a negatively charged region where no trace has been stored. The positively charged areas allow the electrons to pass through to the post accelerator region and the display target phosphor. The negatively charged region repels the flood electrons back to the collector mesh. Thus, the charge pattern on the storage surface appears reproduced on the CRT display phosphor just as though it were being traced with a deflected beam. Figure 5.7 (c) shows a display of the stored charge pattern on a mesh storage.

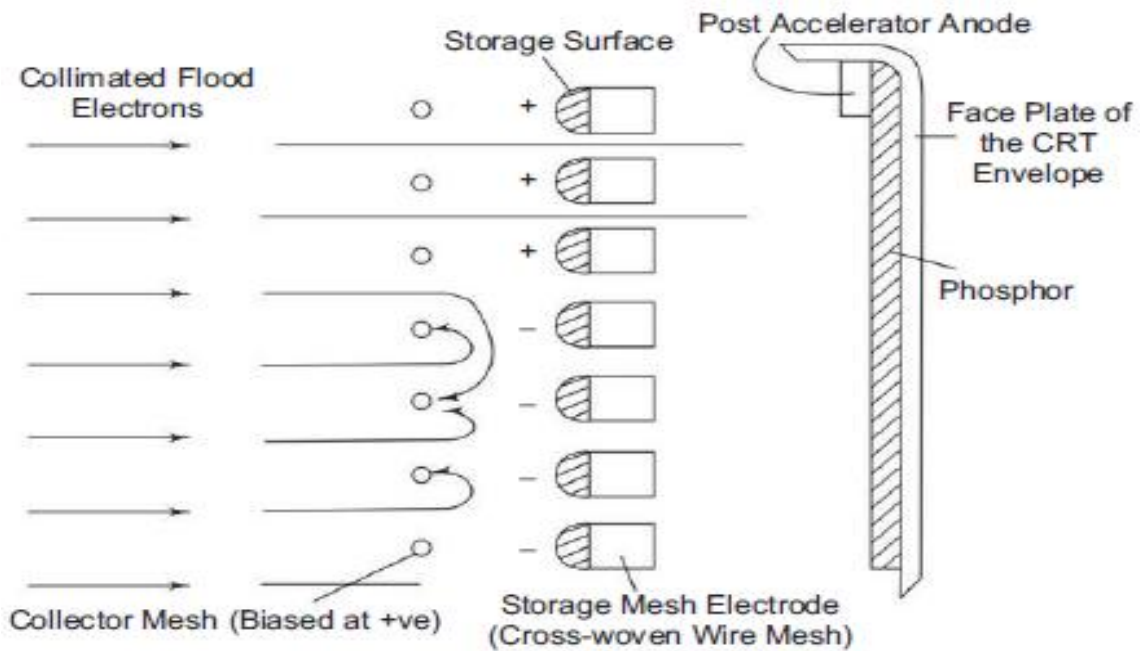


Figure 5.7 (c) Display of stored charged pattern on a mesh-storage

5.8 Standard Signal generator

A standard signal generator produces known and controllable voltages. It is used as power source for the measurement of gain, signal to noise ratio (S/N), bandwidth, standing wave ratio and other properties. It is extensively used in the testing of radio receivers and transmitters. The instrument is provided with a means of modulating the carrier frequency, which is indicated by the dial setting on the front panel. The modulation is indicated by a meter. The output signal can be Amplitude Modulated (AM) or Frequency Modulated (FM). Modulation may be done by a sine wave, square wave, triangular wave or a pulse. The elements of a conventional signal generator are shown in Figure 5.8.

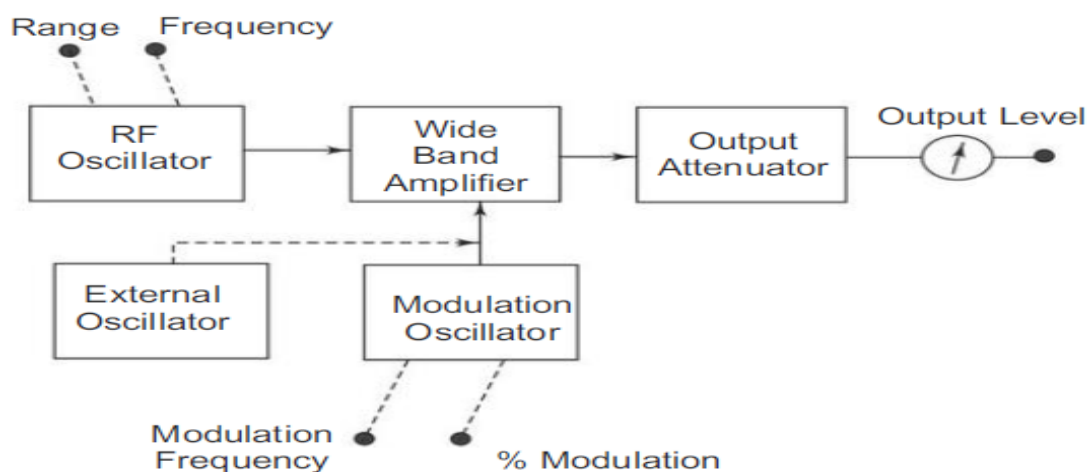


Figure 5.8 Conventional standard signal generator

The carrier frequency is generated by a very stable RF oscillator using an LC tank circuit, having a constant output over any frequency range. The frequency of oscillations is indicated by the frequency range control and the vernier dial setting. AM is provided by an internal sine wave generator or from an external source. (Modulation is done in the output amplifier circuit. This amplifier delivers its output, that is, modulation carrier, to an attenuator. The output voltage is read by an output meter and the attenuator output setting.)

Frequency stability is limited by the LC tank circuit design of the master oscillator. Since range switching is usually accomplished by selecting appropriate capacitors, any change in frequency range upsets the circuit design to some extent and the instrument must be given time to stabilise at the new resonant frequency. In high frequency oscillators, it is essential to isolate the oscillator circuit from the output circuit. This isolation is necessary, so that changes occurring in the output circuit do not affect the oscillator frequency, amplitude and distortion characteristics. Buffer amplifiers are used for this purpose.

5.9 Function Generator

A function generator produces different waveforms of adjustable frequency. The common output waveforms are the sine, square, triangular and sawtooth waves. The frequency may be adjusted, from a fraction of a Hertz to several hundred kHz. The various outputs of the generator can be made available at the same time. For example, the generator can provide a square wave to test the linearity of an amplifier and simultaneously provide a sawtooth to drive the horizontal deflection amplifier of the CRO to provide a visual display. Capability of Phase Lock The function generator can be phase locked to an external source. One function generator can be used to lock a second function generator, and the two output signals can be displaced in phase by adjustable amount.

In addition, the fundamental frequency of one generator can be phase locked to a harmonic of another generator, by adjusting the amplitude and phase of the harmonic, almost any waveform can be generated by addition. The function generator can also be phase locked to a frequency standard and all its output waveforms will then have the same accuracy and stability as the standard source.

The block diagram of a function generator is illustrated in Figure 5.9. Usually, the frequency is controlled by varying the capacitor in the LC or RC circuit. In this instrument the frequency is controlled by varying the magnitude of current which drives the integrator. The instrument produces sine, triangular and square waves with a frequency range of 0.01 Hz to 100 kHz.

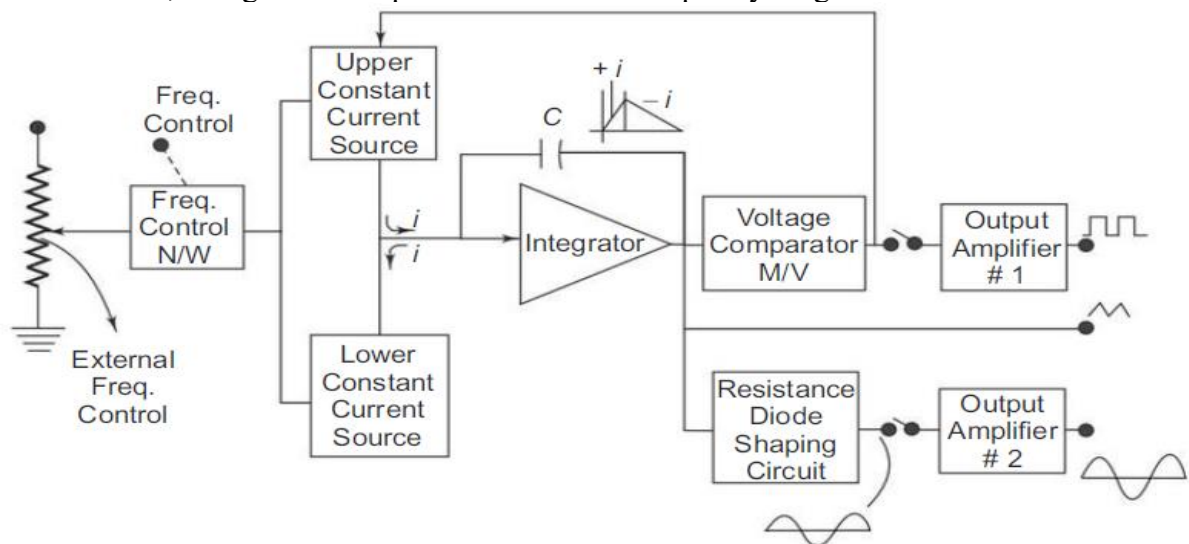


Figure 5.9 Function generator

The frequency-controlled voltage regulates two current sources. The upper current source supplies constant current to the integrator whose output voltage increases linearly with time, according to the equation of the output signal voltage.

$$e_{\text{out}} = -\frac{1}{C} \int_0^t i dt$$

An increase or decrease in the current increases or decreases the slope of the output voltage and hence controls the frequency. The voltage comparator multivibrator changes states at a pre-determined maximum level of the integrator output voltage. This change cuts off the upper current supply and switches on the lower current supply. The lower current source supplies a reverse current to the integrator, so that its output decreases linearly with time. When the output reaches a pre-determined minimum level, the voltage comparator again changes state and switches on the upper current source.

The output of the integrator is a triangular waveform whose frequency is determined by the magnitude of the current supplied by the constant current sources. The comparator output delivers a square wave voltage of the same frequency. The resistance diode network alters the slope of the triangular wave as its amplitude changes and produces a sine wave with less than 1% distortion.

5.10 Random Noise Generator

A simplified block diagram used in the audio frequency range is shown in Figure 5.10.

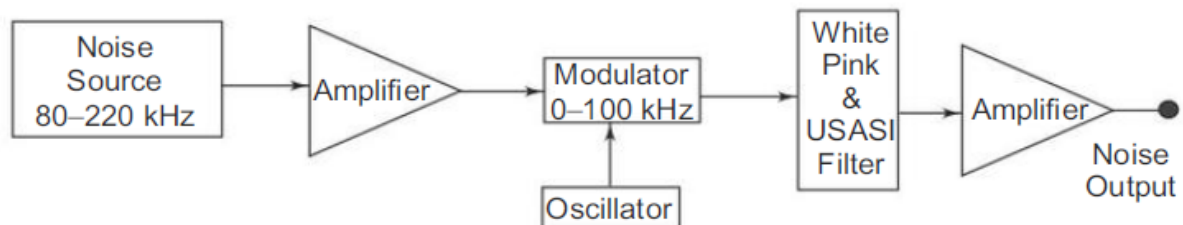


Figure 5.10 Random noise generator

The instrument offers the possibility of using a single measurement to indicate performance over a wide frequency band, instead of many measurements at one frequency at a time. The spectrum of random noise covers all frequencies and is referred to as White noise, i.e. noise having equal power density at all frequencies (an analogy is white light). The power density spectrum tells us how the energy of a signal is distributed in frequency, but it does not specify the signal uniquely, nor does it tell us very much about how the amplitude of the signal varies with time. The spectrum does not specify the signal uniquely because it contains no phase information.

The method of generating noise is usually to use a semiconductor noise diode, which delivers frequencies in a band roughly extending from 80 – 220 kHz. The output from the noise diode is amplified and heterodyned down to the audio frequency band by means of a balanced symmetrical modulator. The filter arrangement controls the bandwidth and supplies an output signal in three spectrum choices, white noise, pink noise and Usasi noise. From Figure 5.11, it is seen that white noise is flat from 20 Hz to 25 kHz and has an upper cutoff frequency of 50 kHz with a cutoff slope of -12 db/octave. Pink noise is so called because the lower frequencies have a larger amplitude, similar to red light. Pink noise has a voltage spectrum which is inversely proportional to the square root of frequency and is used in bandwidth analysis. Usasi noise ranging simulates the energy distribution of speech and music frequencies and is used for testing audio amplifiers and loud speakers.

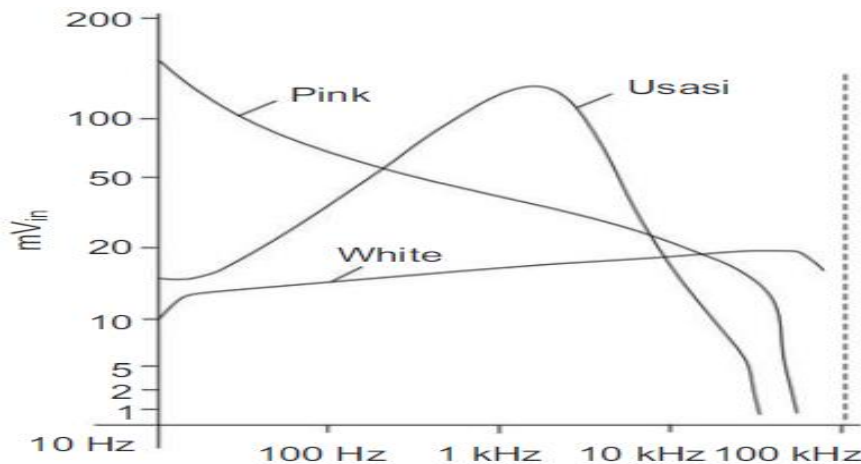


Figure 5.11 Random noise generator

5.12 Basic Wave Analyzer

A basic wave analyzer is shown in Figure 5.12. It consists of a primary detector, which is a simple LC circuit. This LC circuit is adjusted for resonance at the frequency of the particular harmonic component to be measured.

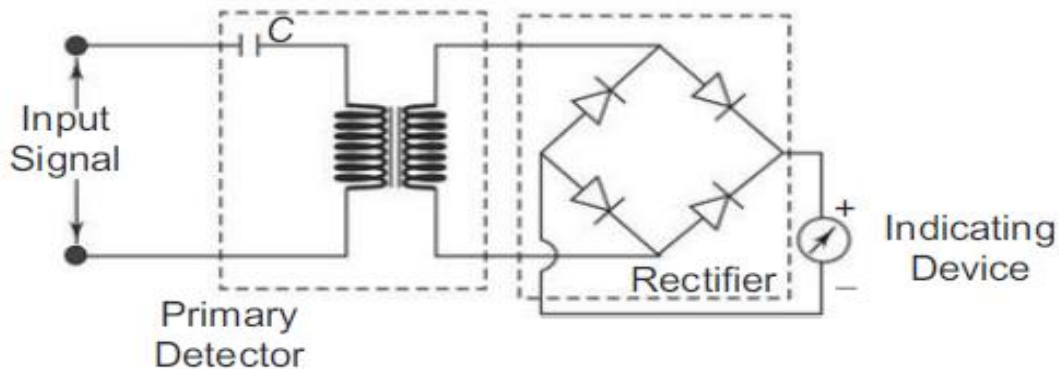


Figure 5.12 Basic wave analyzer

The intermediate stage is a full wave rectifier, to obtain the average value of the input signal. The indicating device is a simple dc voltmeter that is calibrated to read the peak value of the sinusoidal input voltage. Since the LC circuit is tuned to a single frequency, it passes only the frequency to which it is tuned and rejects all other frequencies. A number of tuned filters, connected to the indicating device through a selector switch, would be required for a useful Wave analyzer.

5.13 Heterodyne Wave Analyzer

Wave analyzers are useful for measurement in the audio frequency range only. For measurements in the RF range and above (MHz range), an ordinary wave analyzer cannot be used. Hence, special types of wave analyzers working on the principle of heterodyning (mixing) are used. These wave analyzers are known as Heterodyne wave analyzers. In this wave analyzer, the input signal to be analyzed is heterodyned with the signal from the internal tunable local oscillator in the mixer stage to produce a higher IF frequency. By tuning the local oscillator frequency, various signal frequency components can be shifted within the pass-band of the IF amplifier. The output of the IF amplifier is rectified and applied to the meter circuit. An instrument that involves the principle of heterodyning is the Heterodyning tuned voltmeter, shown in Figure 5.13.

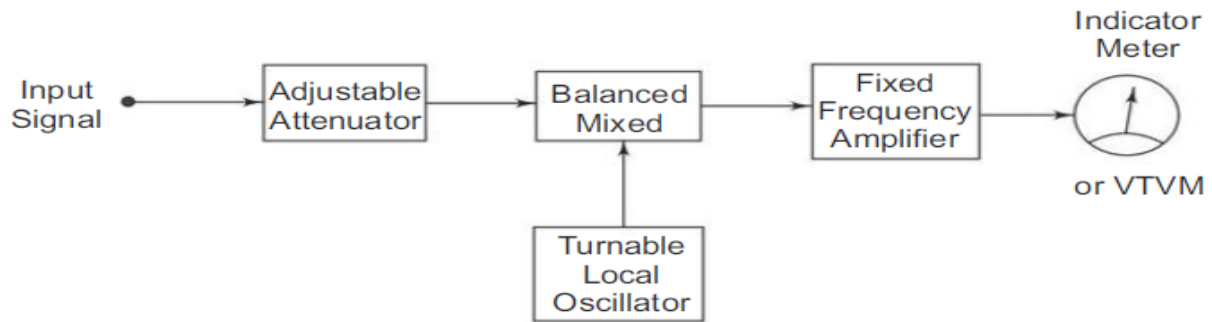


Figure 5.13 Heterodyne wave analyzer

The input signal is heterodyned to the known IF by means of a tunable local oscillator. The amplitude of the unknown component is indicated by the VTVM or output meter. The VTVM is calibrated by means of signals of known amplitude. The frequency of the component is identified by the local oscillator frequency, i.e. the local oscillator frequency is varied so that all the components can be identified. The local oscillator can also be calibrated using input signals of known frequency. The fixed frequency amplifier is a multistage amplifier which can be designed conveniently because of its frequency characteristics. This analyzer has good frequency resolution and can measure the entire AF frequency range. With the use of a suitable attenuator, a wide range of voltage amplitudes can be covered. When a knowledge of the individual amplitudes of the component frequency is desired, a heterodyne wave analyzer is used. A modified heterodyne wave analyzer is shown in Figure 5.14. In this analyzer, the attenuator provides the required input signal for heterodyning in the first mixer stage, with the signal from a local oscillator having a frequency of 30 – 48 MHz.

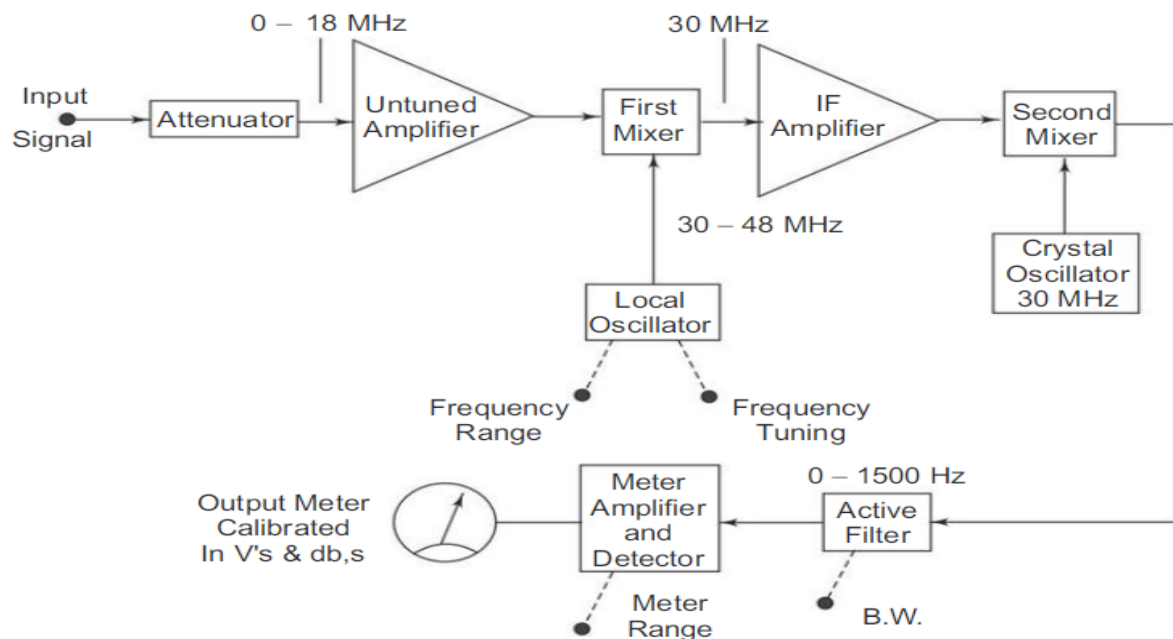


Figure 5.14 RF heterodyne wave analyzer

The first mixer stage produces an output which is the difference of the local oscillator frequency and the input signal, to produce an IF signal of 30 MHz. This IF frequency is uniformly amplified by the IF amplifier. This amplified IF signal is fed to the second mixer stage, where it is again heterodyned to produce a difference frequency or IF of zero frequency. The selected component is then passed to the meter amplifier and detector circuit through an active filter having a controlled band-width. The meter detector output can then be read off on a db-calibrated scale, or may be applied to a secondary device such as a recorder.

This wave analyzer is operated in the RF range of 10 kHz – 18 MHz, with 18 overlapping bands selected by the frequency range control of the local oscillator. The bandwidth, which is controlled by the active filter, can be selected at 200 Hz, 1 kHz and 3 kHz.

5.15 Spectrum Analyzer

The most common way of observing signals is to display them on an oscilloscope, with time as the X-axis (i.e. amplitude of the signal versus time). This is the time domain. It is also useful to display signals in the frequency domain. The instrument providing this frequency domain view is the spectrum analyzer. A spectrum analyzer provides a calibrated graphical display on its CRT, with frequency on the horizontal axis and amplitude (voltage) on the vertical axis. Displayed as vertical lines against these coordinates are sinusoidal components of which the input signal is composed. The height represents the absolute magnitude, and the horizontal location represents the frequency. These instruments provide a display of the frequency spectrum over a given frequency band. Spectrum analyzers use either a parallel filter bank or a swept frequency technique. In a parallel filter bank analyzer, the frequency range is covered by a series of filters whose central frequencies and bandwidth are so selected that they overlap each other, as shown in Figure 5.15 (a). Typically, an audio analyzer will have 32 of these filters, each covering one third of an octave. For wide band narrow resolution analysis, particularly at RF or microwave signals, the swept technique is preferred.

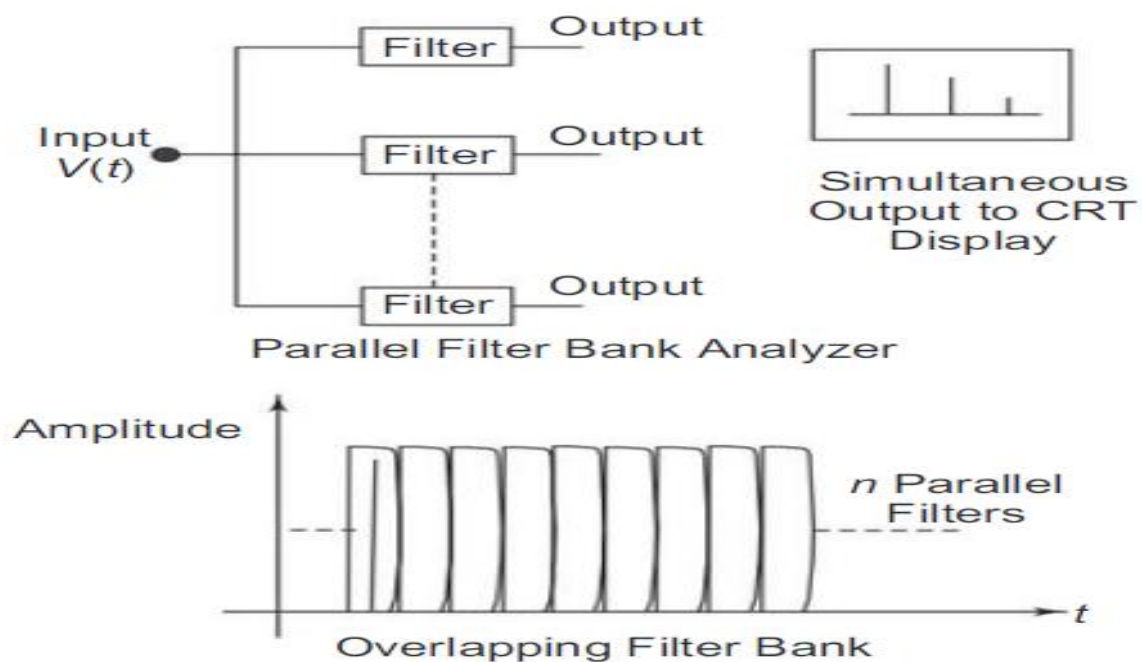


Figure 5.15 (a) Spectrum analyzer (Parallel filter bank analyzer)

5.15.1 Basic Spectrum Analyzer Using Swept Receiver Design

Referring to the block diagram of Figure 5.15 (b), the sawtooth generator provides the sawtooth voltage which drives the horizontal axis element of the scope and this sawtooth voltage is the frequency controlled element of the voltage tuned oscillator. As the oscillator sweeps from $f_{\min}$ to $f_{\max}$ of its frequency band at a linear recurring rate, it beats with the frequency component of the input signal and produce an IF, whenever a frequency component is met during its sweep. The frequency component and voltage tuned oscillator frequency beats together to produce a difference frequency, i.e. IF. The IF corresponding to the component is amplified and detected if necessary, and then applied to the vertical plates of the CRO, producing a display of amplitude versus frequency.

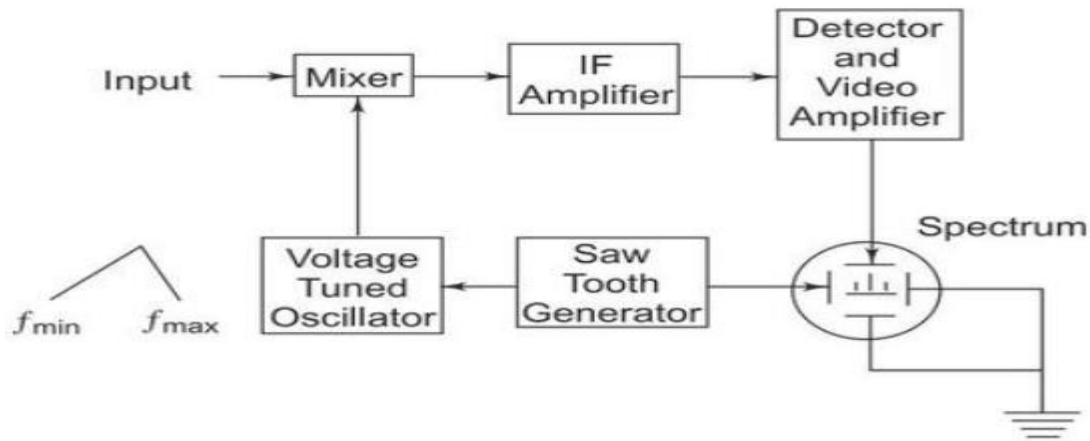


Figure 5.15 (b) Spectrum analyzer

The spectrum produced if the input wave is a single toned A.M. is given in Figures 5.15 (c) and 5.15 (d).

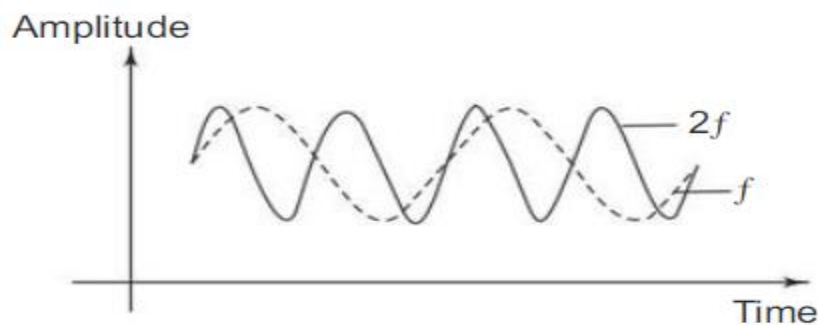


Figure 5.15 (c) Test wave seen on ordinary CRO

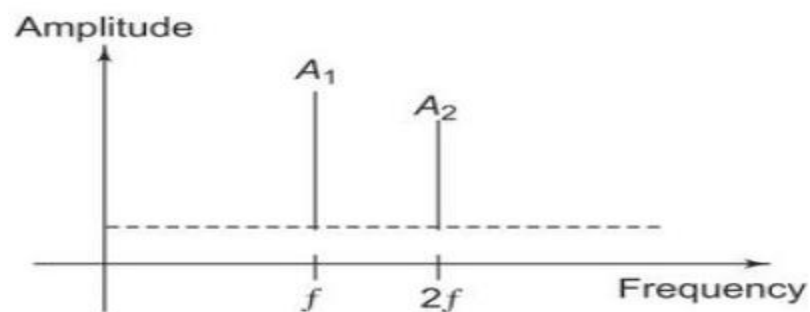


Figure 5.15 (d) Display on the spectrum CRO

One of the principal applications of spectrum analyzers has been in the study of the RF spectrum produced in microwave instruments. In a microwave instrument, the horizontal axis can display as wide a range as 2 – 3 GHz for a broad survey and as narrow as 30 kHz, for a highly magnified view of any small portion of the spectrum. Signals at microwave frequency separated by only a few kHz can be seen individually. The frequency range covered by this instrument is from 1 MHz to 40 GHz. The basic block diagram (Figure 5.15 (e)) is of a spectrum analyzer covering the range 500 kHz to 1 GHz, which is representative of a superheterodyne type.

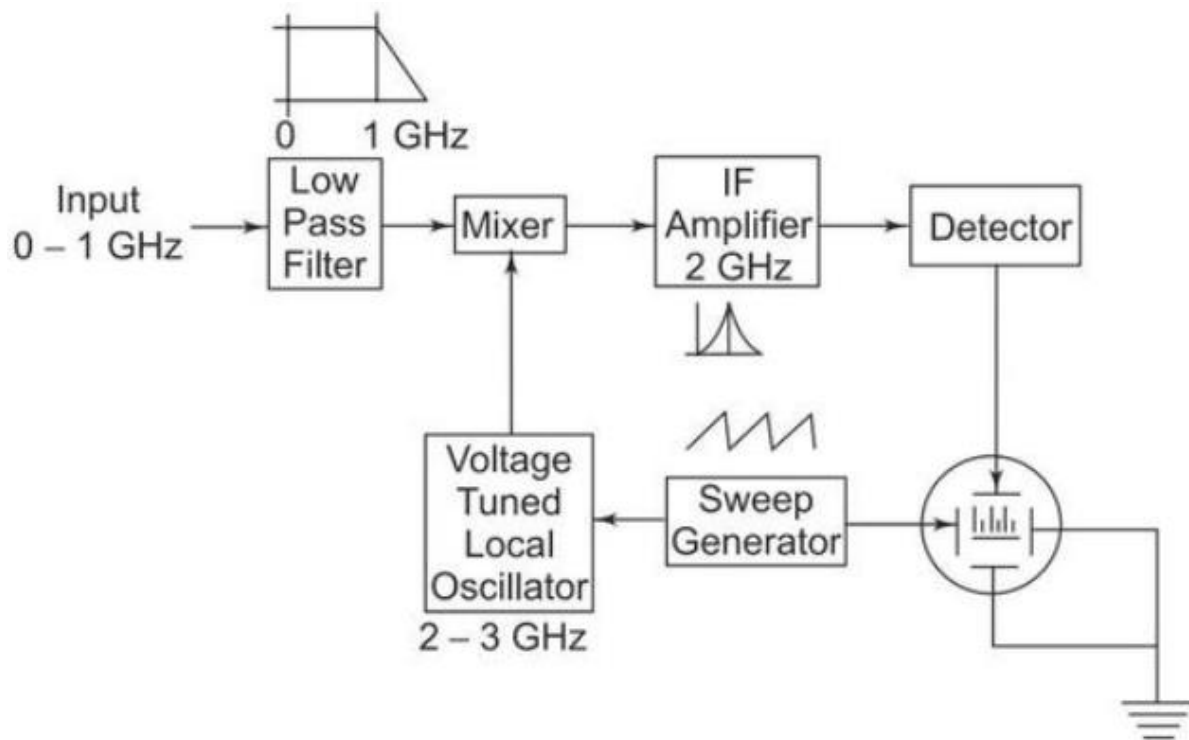


Figure 5.15 (e) RF spectrum analyzer

The input signal is fed into a mixer which is driven by a local oscillator. This oscillator is linearly tunable electrically over the range 2 – 3 GHz. The mixer provides two signals at its output that are proportional in amplitude to the input signal but of frequencies which are the sum and difference of the input signal and local oscillator frequency. The IF amplifier is tuned to a narrow band around 2 GHz, since the local oscillator is tuned over the range of 2 – 3 GHz, only inputs that are separated from the local oscillator frequency by 2 GHz will be converted to IF frequency band, pass through the IF frequency amplifier, get rectified and produce a vertical deflection on the CRT. From this, it is observed that as the sawtooth signal sweeps, the local oscillator also sweeps linearly from 2 – 3 GHz. The tuning of the spectrum analyzer is a swept receiver, which sweeps linearly from 0 to 1 GHz. The sawtooth scanning signal is also applied to the horizontal plates of the CRT to form the frequency axis. (The spectrum analyzer is also sensitive to signals from 4 – 5 GHz referred to as the image frequency of the superheterodyne. A low pass filter with a cutoff frequency above 1 GHz at the input suppresses these spurious signals.) Spectrum analyzers are widely used in radars, oceanography, and bio-medical fields.